

The Interactive Knight's Tour Challenge

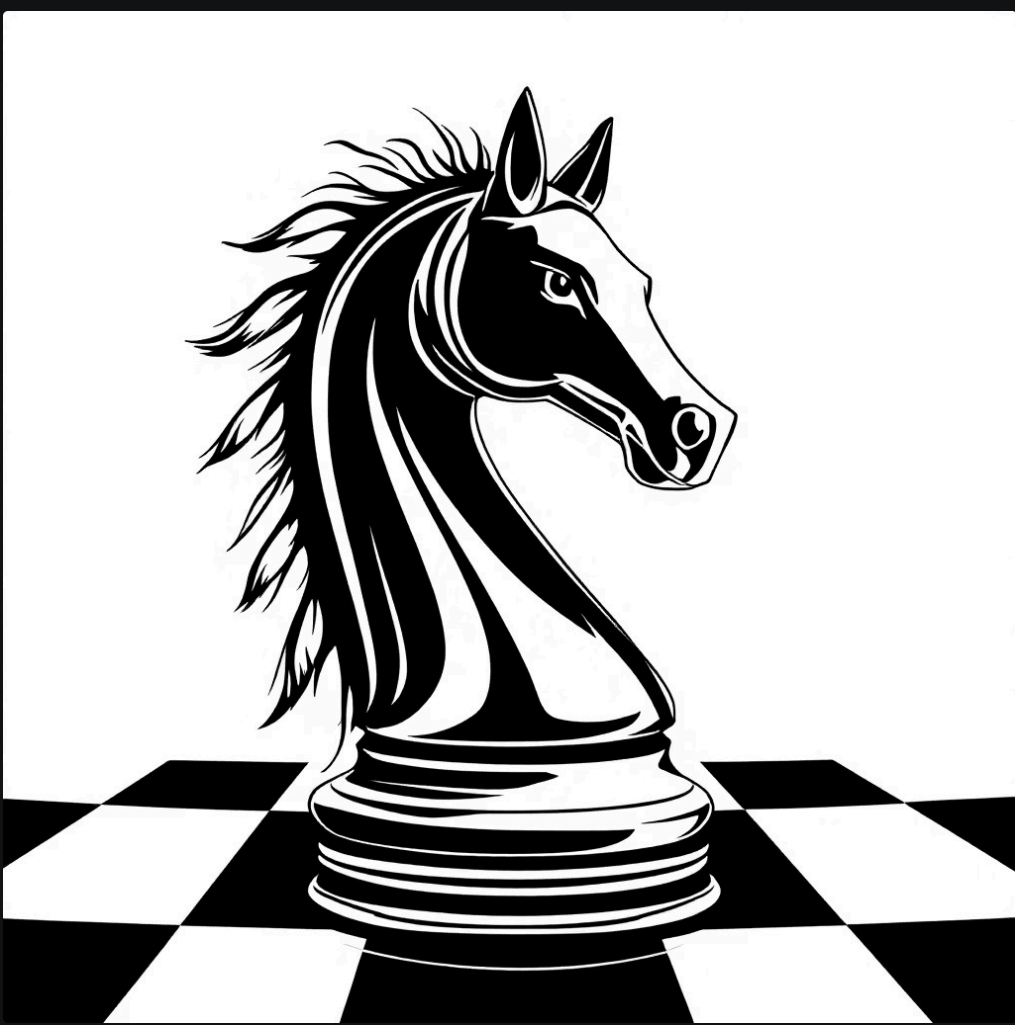
Exploring the fascinating intersection of classical mathematics, chess strategy, and modern web development through an engaging puzzle experience.



Unveiling the Knight's Tour Problem

The Classic Challenge

The Knight's Tour is a timeless mathematical puzzle rooted in chess. The objective is elegantly simple yet deceptively complex: move a knight across a chessboard visiting every square exactly once, following its characteristic L-shaped movement pattern.



Interactive Gameplay

An engaging web application enabling users to tackle the Knight's Tour puzzle with intuitive controls and visual feedback.

Algorithmic Insights

Tools and explanations revealing the computational complexity and sophisticated solving strategies behind this classic problem.

Competitive Spirit

A comprehensive scoring system with real-time leaderboards fostering community engagement and friendly competition.

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Interactive Chessboard Experience

Dynamic board generation supporting multiple sizes (5×5, 6×6) with smooth animations, visual highlighting of visited squares, current position, and all possible next moves.

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Intelligent Gameplay Assistance

Request optimal next-move hints powered by sophisticated algorithms, or query whether completing the tour remains possible from your current position.

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Comprehensive Scoring Mechanism

Rewards for squares visited and hints remaining, with penalties for possibility checks. Difficulty multipliers (Easy, Medium, Hard) adjust final scores accordingly.

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Persistent Leaderboards & Community

Real-time high score tracking across board sizes, with a unique "tourists before you" counter showcasing community participation and engagement.

Responsive Design

Seamlessly optimized across all devices, from smartphones to desktop monitors, ensuring accessibility for every user.

Customisation Options

Personalise your experience with theme selection, difficulty levels with visual board underglow effects, and custom player names.

Informative Guidance

Detailed instructions, algorithmic logic explanations, and transparent scoring rules ensure players understand every aspect of the game.

Technical Architecture: Building Blocks of Excellence

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Frontend Framework

React with TypeScript provides robust, component-based architecture with strong type safety, whilst React Router manages seamless single-page application navigation.

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Styling & UI Components

Tailwind CSS enables rapid custom design development. shadcn/ui components built on Radix UI ensure accessibility. next-themes handles mode switching, whilst lucide-react provides beautiful icons.

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Backend & Database

Supabase powers our backend-as-a-service, leveraging PostgreSQL for data persistence, Row Level Security (RLS) for protection, real-time subscriptions for live updates, and automated functions for intelligent high score management.

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Concurrency Management

Web Workers offload computationally intensive solver operations from the main thread, preventing UI freezing and maintaining silky-smooth responsiveness during complex calculations.

The Algorithm: Intelligence Behind the Knight

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Backtracking (DFS)

Recursive depth-first exploration of possible paths. When encountering dead ends, the algorithm backtracks to previous decision points, guaranteeing solutions when they exist.

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Warnsdorff's Rule

A powerful heuristic optimization: prioritise moves to squares with fewest onward options. By tackling constrained squares early, the knight avoids getting trapped later, dramatically pruning the search tree.

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Web Worker Integration

The entire solving logic executes in a dedicated background thread. Communication via postMessage keeps the UI responsive whilst complex calculations proceed seamlessly in parallel.

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Why Warnsdorff's Rule Works:

By calculating the "degree" (valid next moves) for each potential square and sorting them, the algorithm attempts lower-degree moves first. This intelligent prioritisation, combined with random tie-breaking for solution variety, transforms a potentially exponential search into a remarkably efficient process.

Challenges Conquered & Solutions Implemented

Performance Optimisation for Larger Boards

Challenge: 6×6 boards and beyond demanded intensive computation, risking UI freezes.

Solution: Web Workers isolate solver logic in separate threads, maintaining flawless responsiveness.

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State Management Complexity

Challenge: Tracking board state, knight position, visited squares, path history, and game status across components.

Solution: Strategic use of React's useState and useCallback hooks with clear component hierarchy.

Real-time Leaderboard Synchronisation

Challenge: Ensuring instant leaderboard updates across all active users.

Solution: Supabase real-time PostgreSQL subscriptions automatically refresh scores for all clients.

Player Name Validation

Challenge: Preventing inappropriate entries on public leaderboards.

Solution: Client-side validation against disallowed keywords with immediate user feedback.

Future Enhancements: The Road Ahead

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User Authentication

Integrate Supabase Auth for account creation, progress saving, and personalised high score tracking.

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Expanded Board Options

Custom dimensions and irregular board shapes for enhanced puzzle variety and increased challenge.

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Visual Path Tracing

Animated visualisation of complete solution paths discovered by the solver algorithm.

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Alternative Game Modes

Timed challenges, move limitations, and creative variations to diversify gameplay experiences.

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Performance Analytics

Comprehensive tracking of user metrics, progress patterns, and skill development over time.

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Enhanced AI Solver

Research into advanced heuristics and algorithms for even faster, more sophisticated hint generation.